

# On Regularization of Maximum Mean Discrepancy Gradient Flow

Zonghao Chen   Aratrika Mustafi   Pierre Glaser   Anna Korba  
Arthur Gretton   Bharath K. Sriperumbudur

**Problem:** How to learn a target probability distribution  $\pi$  on  $\mathbb{R}^d$ .

- Sampling (e.g:  $\pi \propto \exp(-V)$  is the posterior distribution in Bayesian inference).
- Generative models (e.g:  $\pi$  is the distribution of an image dataset).

# Background: Optimization in the space of probability measures

**Problem:** How to learn a target probability distribution  $\pi$  on  $\mathbb{R}^d$ .

- This problem can be written as an **optimization** problem on  $\mathcal{P}_2(\mathbb{R}^d)$ .

$$\arg \min_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} D(\mu \mid \pi).$$

- Here  $D$  is a similarity metric or distance, e.g. Kullback–Leibler divergence.
  - $D(\mu \mid \pi) = 0$  if and only if  $\mu = \pi$ .
- $\mathcal{P}_2(\mathbb{R}^d)$  denotes the space of probability measures with a finite second moment.
  - We primarily consider  $\mathcal{P}_2(\mathbb{R}^d)$  rather than  $\mathcal{P}(\mathbb{R}^d)$  for the nice geometrical properties.
- How to find the minimum? Gradient descent!

## Background: Euclidean gradient flow

- Euclidean gradient flow of an objective  $F : \mathbb{R}^d \rightarrow \mathbb{R}$

$$\partial_t x_t = -v(x_t), \quad v = \nabla F.$$

- $\nabla F$  denotes the gradient of  $F$ .
- This is the continuous-time analogue of gradient descent:

$$x_{n+1} = x_n - \gamma v(x_n),$$

where  $\gamma > 0$  is the step size.

- When  $F$  is both **strongly convex and smooth**, Euclidean gradient **descent** converges exponentially fast [BV04].

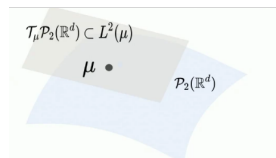
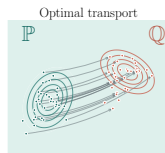
# Background: Wasserstein gradient flow

**Challenge:** How to find  $\arg \min_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} D(\mu \mid \pi)$ ?

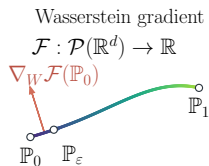
- Gradient descent! Wait... How do we define gradients in  $\mathcal{P}_2(\mathbb{R}^d)$ ?
- Endow  $\mathcal{P}_2(\mathbb{R}^d)$  with the Wasserstein-2 distance  $W_2$ .

$$W_2^2(\nu, \mu) = \int \|T(x) - x\|^2 d\nu(x) = \|T - \text{Id}\|_{L^2(\nu)}^2.$$

- $W_2^2(\nu, \mu)$  means the **minimal energy** takes to transport mass from  $\nu$  to  $\mu$ .
- $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$  is the **optimal transport map** from  $\nu$  to  $\mu$ .
- $(\mathcal{P}_2, W_2)$  can be 'treated' as a Riemann manifold under Otto's calculus [Ott01].
- The **tangent space**  $\mathcal{T}_\mu \mathcal{P}_2(\mathbb{R}^d)$  at  $\mu \in \mathcal{P}_2(\mathbb{R}^d)$  is a dense subset of  $L^2(\mu)$ .



# Background: Wasserstein gradient flow



## Definition (Wasserstein gradient)

Let  $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$ . The **Wasserstein gradient of  $\mathcal{F}$**  evaluated at  $\mu \in \mathcal{P}_2(\mathbb{R}^d)$  is the unique function  $\nabla_{W_2} \mathcal{F}(\mu) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ , s.t. for any  $T \in \mathcal{T}_\mu \mathcal{P}_2(\mathbb{R}^d)$ ,

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\mathcal{F}((\text{Id} + \epsilon T)_\# \mu) - \mathcal{F}(\mu)] = \langle \nabla_{W_2} \mathcal{F}, T \rangle_{L^2(\mu)}.$$

- The gradient is defined along a ‘curve’  $(\text{Id} + \epsilon T)_\# \mu$  in  $\mathcal{P}_2(\mathbb{R}^d)$ .

# Background: Wasserstein gradient flow

## Definition (Wasserstein gradient flow)

**Wasserstein gradient flow** of  $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$  is a sequence of distributions  $(\mu_t)_{t \geq 0}$  that evolves according to the **continuity equation**  $\partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0$ , where the vector field  $v_t = -\nabla_{W_2} \mathcal{F}(\mu_t)$ . Or equivalently,  $\dot{x}_t = v_t(x_t)$  with  $\mu_t = \text{Law}(x_t)$ .

## Euclidean Gradient Flow

- State space:  $\mathbb{R}^d$
- Objective  $F : \mathbb{R}^d \rightarrow \mathbb{R}$ .
- Update scheme:  $\dot{x}_t = v_t(x_t)$ , with  $v_t = -\nabla F$ .

## Wasserstein Gradient Flow

- State space:  $\mathcal{P}_2(\mathbb{R}^d)$
- Objective  $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$ :
- Update scheme  $\dot{x}_t = v_t(x_t)$ , with  $v_t = -\nabla_{W_2} \mathcal{F}(\mu_t)$ .

## Example 1: Langevin diffusion [JKO98]

- Given the target distribution  $\pi \propto \exp(-V)$  with  $V : \mathbb{R}^d \rightarrow \mathbb{R}$ .
- The functional  $\mathcal{F}_{\text{KL}} = \text{KL}(\cdot \| \pi)$  and its Wasserstein gradient

$$[\nabla_{W_2} \mathcal{F}_{\text{KL}}(\mu)](\cdot) = \nabla V(\cdot) + \nabla \log \mu_t(\cdot).$$

- The Wasserstein gradient flow of  $\mathcal{F}_{\text{KL}}$

$$\partial_t \mu_t = \nabla \cdot (\mu_t (\nabla V + \nabla \log \mu_t)).$$

- It is equivalent to the Fokker–Planck equation of the **Langevin diffusion** [SS19]:

$$\mathrm{d}x_t = -\nabla V(x_t) \mathrm{d}t + \sqrt{2} \mathrm{d}W_t, \quad \mu_t = \text{Law}(x_t).$$

- A standard time-discretization (Euler–Maruyama scheme) is:

$$x_{n+1} = x_n - \gamma \nabla V(x_n) + \sqrt{2\gamma} \xi_n, \quad \xi_n \sim \mathcal{N}(0, I_d).$$

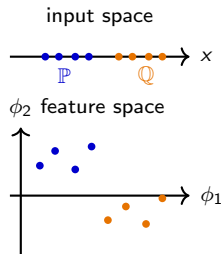


# Background: Maximum Mean Discrepancy (MMD) [GBR<sup>+</sup>12]

- $\mathcal{H}$  is a reproducing kernel Hilbert space (RKHS) with kernel  $k$ .

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- The RKHS contains functions that are linear combinations of the feature maps  $\phi : \mathcal{X} \rightarrow \mathcal{H}$ ,  $f = \sum_{i=1}^n \alpha_i \phi(x_i)$ .



MMD admits a closed-form expression

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

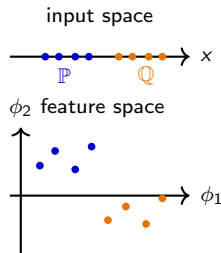
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## Example 2: MMD gradient flow [AKSG19]

- Given  $M$  i.i.d samples  $\{y_i\}_{i=1}^M$  from a target distribution  $\pi$ .
- The functional  $\mathcal{F}_{\text{MMD}} = \frac{1}{2}\text{MMD}^2(\cdot\|\pi)$
- Its Wasserstein gradient

$$\begin{aligned} [\nabla_{W_2}\mathcal{F}_{\text{MMD}}(\mu)](\cdot) &= \nabla \left( \int k(x, \cdot) \, d\mu(x) - \int k(x, \cdot) \, d\pi(x) \right) \\ &\approx \int \nabla_2 k(x, \cdot) \, d\mu(x) - \frac{1}{M} \sum_{i=1}^M \nabla_2 k(y_i, \cdot). \end{aligned}$$

- The Wasserstein gradient flow of  $\mathcal{F}_{\text{MMD}}$ ,

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_t)), \quad dx_t = -[\nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_t)](x_t) dt.$$

- A standard time-discretization (Euler–Maruyama scheme) is:

$$x_{n+1} = x_n - \gamma \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_n)(x_n), \quad \text{Law}(x_n) = \mu_n.$$

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# Convexity and Smoothness

## Wasserstein gradient flow of $\mathcal{F}_{\text{KL}}$

- $\pi \propto \exp(-V)$
- Sampling
- When  $\pi \propto \exp(-V)$  is **strongly log-concave**, i.e.,  $\mathbf{H}V \geq \alpha \text{Id}$ , then  $\mathcal{F}_{\text{KL}}$  is  **$\alpha$ -convex**.
- Convergence in discrete time

$$\mathcal{F}_{\text{KL}}(\mu_n \| \pi) \leq \exp(-\alpha\gamma n) \mathcal{F}_{\text{KL}}(\mu_0 \| \pi) + \frac{\gamma n \beta^2}{\alpha}.$$

- $\beta$  is the Lipschitz continuity of  $V$ .
- It takes  $\mathcal{O}(\frac{1}{\alpha\delta} \log \frac{1}{\delta})$  to reach  $\delta$  error.

## Wasserstein gradient flow of $\mathcal{F}_{\text{MMD}}$

- $\{y_i\}_{i=1}^M$  i.i.d samples from  $\pi$
- Generative modelling
- When  $k$  is bounded and has bounded derivatives, then  $\mathcal{F}_{\text{MMD}}$  is  **$M$ -smooth and  $-M$ -convex**.
- **Non-convergence** in discrete time

$$\mathcal{F}_{\text{MMD}}(\mu_n, \pi) \leq \frac{W_2^2(\mu_0, \pi)}{\gamma n} + \bar{K}.$$

- $\bar{K}$  is a positive barrier term that does not vanish.
- $\lim_{n \rightarrow \infty} \mathcal{F}_{\text{MMD}}(\mu_n, \pi) \neq 0!$

# Non-convexity of MMD prevents global convergence

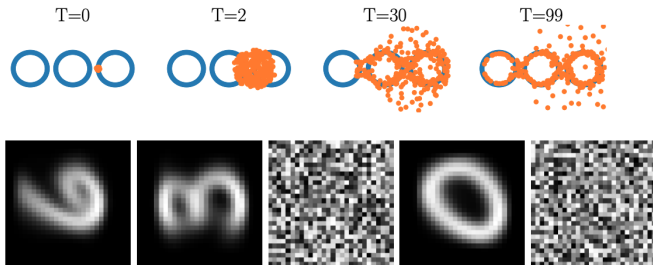


Figure: [BSM25]

- Convergence of MMD flow has been proved under noise injection and stringent conditions on the noise scale [AKSG19, CKK<sup>+</sup>25].
- Disclaimer: There exists many papers where MMD flow **empirically** generate high-quality images with adversarial training of kernels [GBG25], or with non-smooth kernels plus deep network distillation [HWAH24, AHS23].

# Convexity and Smoothness

**Question:** Can we find a new objective  $\mathcal{F}$  such that it enjoys **convergence**, similar to  $\mathcal{F}_{\text{KL}}$ , in the **generative modelling** setting where only samples are available?

## Proposition 1 (MMD and $\chi^2$ -divergence)

Suppose  $\mu$  is absolutely continuous with respect to  $\pi$ , i.e.,  $\mu \ll \pi$ . Then

$$\text{MMD}^2(\mu \parallel \pi) = \left\| T_{\pi}^{\frac{1}{2}} \left( \frac{d\mu}{d\pi} - 1 \right) \right\|_{L^2(\pi)}^2 \quad \text{and} \quad \chi^2(\mu \parallel \pi) = \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)}^2.$$

Here,  $T_{\pi} : L^2(\pi) \rightarrow L^2(\pi)$  is the kernel **integral operator** defined as

$$T_{\pi} f(\cdot) = \int k(x, \cdot) f(x) \, d\pi(x).$$

- Similar to  $\mathcal{F}_{\text{KL}}$  flow,  $\mathcal{F}_{\chi^2}(\cdot) = \chi^2(\cdot \parallel \pi)$  flow enjoys convergence for Gaussian-like  $\pi$  [OT11].
- An **interpolation** between  $\text{MMD}^2$  and  $\chi^2$ ?

# DrMMD: An interpolation of MMD and $\chi^2$ -divergence

## Definition (De-regularized Maximum Mean Discrepancy (DrMMD))

Suppose  $\mu \ll \pi$  where  $\mu, \pi \in \mathcal{P}_2(\mathbb{R}^d)$ . Then the (de)-regularized maximum mean discrepancy (DrMMD) between  $\mu, \pi$  is defined as, for  $\lambda > 0$ ,

$$\text{DrMMD}(\mu \parallel \pi) = (1 + \lambda) \left\| \left( (T_\pi + \lambda \text{Id})^{-1} T_\pi \right)^{\frac{1}{2}} \left( \frac{d\mu}{d\pi} - 1 \right) \right\|_{L^2(\pi)}^2.$$

## Proposition 2 (Interpolation of $\text{MMD}^2$ and $\chi^2$ )

*Suppose  $k$  is bounded, continuous, and  $c_0$ -universal.*

$$\lim_{\lambda \rightarrow 0} \text{DrMMD}(\mu \parallel \pi) = \chi^2(\mu \parallel \pi), \quad \lim_{\lambda \rightarrow \infty} \text{DrMMD}(\mu \parallel \pi) = \text{MMD}^2(\mu \parallel \pi).$$

- Similar ideas based on spectral regularization have been used for kernel hypothesis testing [MRW<sup>+</sup>99, HBM07, HKSL24].
- Interpolation between MMD and general f-divergences [GAG21, SNRS26].



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# DrMMD: An interpolation of MMD and $\chi^2$

**Question:** Does DrMMD inherit the advantages of MMD<sup>2</sup> and  $\chi^2$ ?

- Does  $\mathcal{F}_{\text{DrMMD}}$  flow admit **finite sample implementation** of its Wasserstein gradient flow?
- Does  $\mathcal{F}_{\text{DrMMD}}$  flow **convergence** when  $\pi$  is Gaussian-like?

# Kernel assumptions

## Assumption 1 (Growth I)

*There exists a constant  $M > 0$  such that, for every  $x \in \mathbb{R}^d$ ,*

$$\begin{aligned} |k(x, x)| &\leq M (1 + |x|^4), \\ |\partial_{1,i} \partial_{2,i} k(x, x)| &\leq M (1 + |x|^2), \quad i \in [d]. \end{aligned}$$

## Assumption 2 (Local Boundedness)

*For  $R > 0$ , let  $\mathcal{K} = B(0, R) \subset \mathbb{R}^d$ . There exists a constant  $B_R > 0$  such that, for every  $i, j \in [d]$ ,*

$$\sup_{x \in \mathcal{K}} \|\partial_{1,i} \partial_{1,j} k(x, \cdot)\|_{\mathcal{H}} \leq B_R.$$

- Gaussian, Matérn, and inverse multiquadratic kernels.

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## Proposition 3 (Finite-sample estimate of the Wasserstein gradient of $\mathcal{F}_{\text{DrMMD}}$ )

The Wasserstein gradient of  $\mathcal{F}_{\text{DrMMD}}(\cdot) = \text{DrMMD}(\cdot \| \pi)$  at  $\mu$  is  $(1 + \lambda) \nabla h_{\mu, \pi}(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ , where

$$h_{\mu, \pi} = (T_{\pi} + \lambda I)^{-1} T_{\pi} \left( \frac{d\mu}{d\pi} - 1 \right).$$

Given empirical distributions  $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$  and  $\hat{\pi} = \frac{1}{M} \sum_{i=1}^M y_i$ . Given the Gram matrices  $K_{xx} \in \mathbb{R}^{N \times N}$ ,  $K_{yy} \in \mathbb{R}^{M \times M}$ ,  $K_{xy} \in \mathbb{R}^{N \times M}$ .

$$\begin{aligned} h_{\hat{\mu}, \hat{\pi}}(\cdot) &= \frac{1}{N\lambda} k(\cdot, x_{1:N}) \mathbb{1}_N - \frac{1}{M\lambda} k(\cdot, y_{1:M}) \mathbb{1}_M - \frac{1}{M\lambda} k(\cdot, y_{1:M}) (M\lambda I + K_{yy})^{-1} K_{yx} \mathbb{1}_N \\ &\quad + \frac{1}{M\lambda} k(\cdot, y_{1:M}) (M\lambda I + K_{yy})^{-1} K_{yy} \mathbb{1}_M. \end{aligned}$$

# Finite-sample implementation

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$$\begin{aligned} h_{\hat{\mu}, \hat{\pi}}(\cdot) = & \frac{1}{N\lambda} k(\cdot, x_{1:N}) \mathbb{1}_N - \frac{1}{M\lambda} k(\cdot, y_{1:M}) \mathbb{1}_M - \frac{1}{M\lambda} k(\cdot, y_{1:M}) (M\lambda I + K_{yy})^{-1} K_{yx} \mathbb{1}_N \\ & + \frac{1}{M\lambda} k(\cdot, y_{1:M}) (M\lambda I + K_{yy})^{-1} K_{yy} \mathbb{1}_M. \end{aligned}$$

# Convergence

## Definition (Poincaré inequality)

We say that  $\pi$  satisfies a **Poincaré inequality** with constant  $C_P$  if for all  $f, \nabla f \in L^2(\pi)$ ,

$$\mathrm{Var}_\pi[f] \leq C_P \mathbb{E}_\pi [\|\nabla f\|^2] .$$

- By “Gaussian-like,” we mean that  $\pi$  satisfies a Poincaré inequality, which in particular implies sub-exponential tails [BGL<sup>+</sup>14].
- For example, if  $\pi = \mathcal{N}(0, a \cdot \mathrm{Id})$ , then  $C_P = a^{-1}$ .

## Proposition 4 (Convergence of $\mathcal{F}_{\chi^2}$ flow)

*Suppose that  $\pi$  satisfies the Poincaré inequality. Let  $(\mu_t)_{t \geq 0}$  follow the  $\mathcal{F}_{\chi^2}$  flow. Then, for any  $T \geq 0$ ,  $\mathrm{KL}(\mu_T \| \pi) \leq \exp(-2C_P^{-1}T) \mathrm{KL}(\mu_0 \| \pi)$ .*

# Convergence of DrMMD gradient flow

## Theorem 1 (Convergence of DrMMD gradient flow)

Let  $(\mu_t)_{t \geq 0}$  be the Wasserstein gradient flow of  $\mathcal{F}_{\text{DrMMD}}$ . Suppose the target distribution  $\pi$  satisfies a Poincaré inequality with constant  $C_P$ . Suppose that  $\frac{d\mu_t}{d\pi} - 1 \in \text{Ran}(T_\pi^r)$  with  $r > 0$ . Then for any  $T \geq 0$ ,

$$\text{KL}(\mu_T \| \pi) \leq \exp\left(-\frac{2(1+\lambda)}{C_P} T\right) \text{KL}(\mu_0 \| \pi) + \mathcal{O}(\lambda^r C_P).$$

- When  $\lambda = 0$ , it recovers the exponential convergence of  $\chi^2$  flow.

$$\text{KL}(\mu_T \| \pi) \leq \exp\left(-\frac{2}{C_P} T\right) \text{KL}(\mu_0 \| \pi).$$

- Larger  $r$  means more regularity of the trajectory and thus smaller bias.
- Smaller Poincaré means faster convergence.
- For continuous time DrMMD flow, we would want  $\lambda \rightarrow 0$  for 'convexity', however, that is not the case for discrete time flow.



# Interpolation space $\text{Ran}(T_\pi^r)$

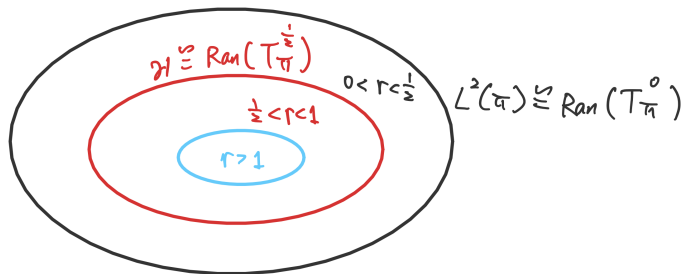


Figure: Visualization of  $\text{Ran}(T_\pi^r)$ .

- Large  $r$  indicates larger smoothness.
- $\text{Ran}(T_\pi^0) \cong L^2(\pi)$  and  $\text{Ran}(T_\pi^{\frac{1}{2}}) \cong \mathcal{H}$ .

- DrMMD gradient flow (continuity equation)

$$\partial_t \mu_t + \nabla \cdot (\mu_t (1 + \lambda) \nabla h_{\mu_t, \pi}) = 0$$

- DrMMD gradient descent: for step size  $\gamma > 0$ ,

$$\mu_{n+1} = (\text{Id} + \gamma(1 + \lambda) \nabla h_{\mu_n, \pi})_{\#} \mu_n.$$

# Convergence of DrMMD gradient descent

## Proposition 5 (Descent lemma of DrMMD gradient descent)

Let  $(\mu_n)_{n \in \mathbb{N}}$  be the Wasserstein gradient descent of  $\mathcal{F}_{\text{DrMMD}}$ . Suppose  $\pi \propto \exp(-V)$  with  $\mathbf{H}V \preceq \beta \mathbf{I}$ . Suppose all assumptions in the proposition of the PL condition on  $\mathcal{F}_{\text{DrMMD}}$  hold. Suppose the step size  $\gamma$  is small enough.

$$\text{KL}(\mu_{n+1} \parallel \pi) - \text{KL}(\mu_n \parallel \pi) \leq -\frac{2}{C_P} \chi^2(\mu_n \parallel \pi) \gamma + \underbrace{\gamma \lambda^r}_{\text{Approximation error}} + \underbrace{\gamma^2 \beta \chi^2(\mu_n \parallel \pi) \frac{1}{\lambda}}_{\text{Discretization error}}$$

- A **trade-off** between the approximation error and the time-discretization error.
- Optimal choice of **adaptive**  $\lambda_n$  at each iterate  $n$ :

$$\lambda_n \propto \chi^2(\mu_n \parallel \pi)^{\frac{1}{r+1}}$$

- At the start, we want a **larger**  $\lambda$  to have more smoothness; when closer to the convergence, we want a **smaller**  $\lambda$  to operate in the  $\chi^2$  regime to better catch the difference.

# Convergence of DrMMD gradient descent

## Theorem 2 (Convergence of DrMMD gradient descent)

Let  $(\mu_n)_{n \in \mathbb{N}}$  be the Wasserstein gradient descent of  $\mathcal{F}_{\text{DrMMD}}$ . Suppose all conditions from the descent lemma hold. Then, for any  $n_{\max} \in \mathbb{N}$ ,

$$\text{KL}(\mu_{n_{\max}} \parallel \pi) \leq \exp\left(-\frac{2n_{\max}\gamma}{C_P}\right) \text{KL}(\mu_0 \parallel \pi) + \mathcal{O}(\gamma^{\frac{r}{r+1}} C_P \beta^{\frac{r}{r+1}}).$$

- To reach error  $\text{KL}(\mu_{n_{\max}} \parallel \pi) \leq \delta$ , it takes  $\mathcal{O}((\frac{1}{\delta})^{\frac{r+1}{r}} \log \frac{1}{\delta})$  iterations.
- By comparison, for Langevin Monte Carlo, it takes  $\mathcal{O}(\frac{1}{\delta} \log \frac{1}{\delta})$  [CEL<sup>+</sup>24].
- DrMMD gradient descent takes more iterations due to the additional approximation error  $\mathcal{O}(\lambda^r)$ , but it **operates without the knowledge of potential  $V$**  and only requires  $\Sigma_\pi = \int k(x, \cdot) \otimes k(x, \cdot) \, d\pi$  and the embedding  $\int k(x, \cdot) \, d\pi$ .

# Conclusions

- We propose DrMMD gradient flow as an interpolation of MMD gradient flow and  $\chi^2$  gradient flow.
- DrMMD gradient flow / descent has **global convergence** results, compared to MMD flow, under an adaptive regularization parameter  $\lambda$ .
- This justifies the application of **adaptive kernels** in recent MMD flow (MMD GAN) papers that achieve SOTA empirical performances.
- More in the paper <https://arxiv.org/pdf/2409.14980>.
  - Empirical results on synthetic datasets.
  - An example of DrMMD flow with a Gaussian target distribution  $\pi$  which satisfies all conditions in the theorems.
  - Finite-particle convergence results with propagation of chaos bound.

# About Me



- Zonghao Chen
- 4th year PhD Student at University College London (UCL)
  - Foundational AI Centre
  - Gatsby Computational Neuroscience Unit
- Graduated from Tsinghua University in 2022
  - Department of EE
- Kernel (nonparametric) methods, causal inference, statistical learning theory

- [AHS23] Fabian Altekrüger, Johannes Hertrich, and Gabriele Steidl. Neural wasserstein gradient flows for maximum mean discrepancies with riesz kernels. *arXiv preprint arXiv:2301.11624*, 2023.
- [AKSG19] Michael Arbel, Anna Korba, Adil Salim, and Arthur Gretton. Maximum mean discrepancy gradient flow. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.
- [BGL<sup>+</sup>14] Dominique Bakry, Ivan Gentil, Michel Ledoux, et al. *Analysis and geometry of Markov diffusion operators*, volume 103. Springer, 2014.
- [BSM25] Ayoub Belhadji, Daniel Sharp, and Youssef Marzouk. Weighted quantization using mmd: From mean field to mean shift via gradient flows. *arXiv preprint arXiv:2502.10600*, 2025.
- [BV04] Stephen P. Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.
- [CEL<sup>+</sup>24] Sinho Chewi, Murat A. Erdogdu, Mufan Li, Ruoqi Shen, and Matthew S. Zhang. Analysis of Langevin Monte Carlo from Poincare to log-Sobolev. *Foundations of Computational Mathematics*, pages 1–51, 2024.

- [CKK<sup>+</sup>25] Zonghao Chen, Toni Karvonen, Heishiro Kanagawa, François-Xavier Briol, and Chris Oates. Stationary MMD points for cubature. 2025.
- [GAG21] Pierre Glaser, Michael Arbel, and Arthur Gretton. KALE flow: A relaxed KL gradient flow for probabilities with disjoint support. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8018–8031. Curran Associates, Inc., 2021.
- [GBG25] Alexandre Galashov, Valentin De Bortoli, and Arthur Gretton. Deep MMD gradient flow without adversarial training. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [GBR<sup>+</sup>12] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(1):723–773, 2012.
- [HBM07] Zaïd Harchaoui, Francis Bach, and Eric Moulines. Testing for homogeneity with kernel Fisher discriminant analysis. In J. Platt, D. Koller, Y. Singer, and S. Roweis, editors, *Advances in Neural Information Processing Systems*, volume 20. Curran Associates, Inc., 2007.



- [HKSL24] Omar Hagrass, Bharath K. Sriperumbudur, and Bing Li. Spectral regularized kernel two-sample tests. *The Annals of Statistics*, 52(3):1076–1101, 2024.
- [HWAH24] Johannes Hertrich, Christian Wald, Fabian Altekrüger, and Paul Hagemann. Generative sliced MMD flows with riesz kernels. In *The Twelfth International Conference on Learning Representations*, 2024.
- [JKO98] Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998.
- [MRW<sup>+</sup>99] Sebastian Mika, Gunnar Rätsch, Jason Weston, Bernhard Schölkopf, and Klaus-Robert Müller. Fisher discriminant analysis with kernels. In *Neural Networks for Signal Processing IX: Proceedings of the 1999 IEEE Signal Processing Society Workshop (Cat. No.98TH8468)*, pages 41–48, 1999.
- [OT11] Shin-ichi Ohta and Asuka Takatsu. Displacement convexity of generalized relative entropies. *Advances in Mathematics*, 228(3):1742–1787, 2011.
- [Ott01] Felix Otto. The geometry of dissipative evolution equations: the porous medium equation. 2001.

- [SFL11] Bharath K. Sriperumbudur, Kenji Fukumizu, and Gert R. G. Lanckriet. Universality, characteristic kernels and RKHS embedding of measures. *Journal of Machine Learning Research*, 12(7):2389–2410, 2011.
- [SNRS26] Viktor Stein, Sebastian Neumayer, Nicolaj Rux, and Gabriele Steidl. Wasserstein gradient flows for moreau envelopes of f-divergences in reproducing kernel hilbert spaces. *Analysis and Applications*, 24(01):21–65, 2026.
- [SS19] Simo Särkkä and Arno Solin. *Applied stochastic differential equations*, volume 10. Cambridge University Press, 2019.